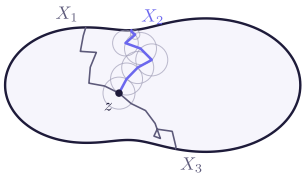
(a) Reuse geometric work

Independent randomizations



$X_i = (U_i, P_i)$

Primal and adjoint may share the walk

(b) Hold out the matching adjoint

Example: use the second primal

pair	primal	adjoint
X_1	$U_1 - h_z$	P_1
X_2	$U_2 - h_z$	~~P_2~~ held out
X_3	$U_3 - h_z$	P_3

$$\bar{P}_{-2} = \frac{P_1 + P_3}{2}$$

(c) Multiply, then average

Two independent factors per term

$$C_2 = -(U_2 - h_z)\bar{P}_{-2}$$

Repeat this holdout for every pair

$$\widehat{D}_T = \frac{1}{N}\sum_{i=1}^{N} C_i$$

$$\bar{P}_{-i} = \frac{S_P - P_i}{N - 1}$$

One adjoint sum $S_P = \sum_j P_j$ $O(N)$ total work N independent pairs; products share samples